FAIZAN MANSHAD (FA22-CVE-051)

SUMMARY

This project involved the hydraulic design and performance evaluation of a combined sewerage system for a 102-hectare urban catchment located near Markazi Jamia Masjid, Mall Road, Wah Cantt. The objective was to develop a reliable, efficient, and code-compliant system to manage both wastewater and stormwater, while resolving existing problems of surcharging, flooding, and inadequate drainage capacity. The system was modeled and optimized using **EPA-SWMM** in accordance with WASA and PHED Punjab Design Guidelines.

The design adopted RCC pipes with a Manning's roughness coefficient of **n = 0.013**. A 20-year design population of **3,835 persons** was projected using geometric growth, and the peak wastewater flow of **15.7 L/s** was calculated using Harmon's peak factor. System performance was evaluated under two rainfall scenarios: a **30 mm/hr** typical storm and a **60 mm/hr** extreme monsoon event, representing critical design conditions.

Key Engineering Challenge: Steep and Undulating Terrain

The most significant challenge encountered during the design was the steep natural topography, with ground elevations falling from approximately *1590 ft* to *1543 ft* across the catchment. This created a conflict between velocity control and self-cleansing requirements. In steep sections, excessive pipe slopes resulted in flow velocities exceeding the allowable limit of ***7 ft/s***, posing a serious risk of scouring and structural damage to RCC conduits. Conversely, in flatter areas, reducing slopes to control velocity risked lowering flow velocities below the minimum self-cleaning requirement of ***2.0 ft/s***, increasing the likelihood of sedimentation and blockage.

Initial EPA-SWMM simulations under the 60 mm/hr rainfall event confirmed these issues, showing excessive velocities, conduit surcharging, and surface flooding at multiple locations.

Design Optimization and Final Outcome

To resolve these challenges, a systematic hydraulic optimization strategy was implemented. Drop manholes were introduced at locations where invert level differences exceeded 3 ft to dissipate excess energy. **Pipe slopes and diameters were carefully adjusted to balance hydraulic efficiency and velocity control.** Following optimization, all conduits achieved flow velocities within the acceptable range of ***2.0 to 7.0 ft/s***, and surcharge and flooding conditions were eliminated under both storm scenarios.

Conclusion

The finalized design successfully addresses the complex hydraulic demands imposed by the steep terrain while complying with all relevant design standards. The optimized combined sewerage system provides a safe, sustainable, and hydraulically stable solution for Wah Cantt, ensuring reliable wastewater conveyance and effective stormwater management under both normal and extreme rainfall conditions.